

States of Matter

Solid, Liquid, and Gas -- An Accessible Learning Guide

This guide explains the three common states of matter in plain language. Every diagram in this document is followed by a written description, so the same information is available whether or not the diagram itself can be seen. Headings are structured for easy navigation, and text and background colors are chosen for strong contrast and readability.

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1. What is Matter?

Matter is anything that takes up space and has mass (weight caused by gravity). Your desk, the air you breathe, and the water in a glass are all matter. Matter is made of tiny particles -- so small that no one can see a single one without a very powerful microscope.

Matter usually exists in one of three everyday states: **solid**, **liquid**, and **gas**. What state matter is in depends on two things: how closely packed its particles are, and how much energy (movement) those particles have.

As a general rule: the more energy particles have, the faster they move and the further apart they spread. Solids have the least particle energy and movement; gases have the most.

2. Solids

In a solid, particles are packed tightly together in a fixed, orderly pattern. They vibrate in place but do not move past one another.

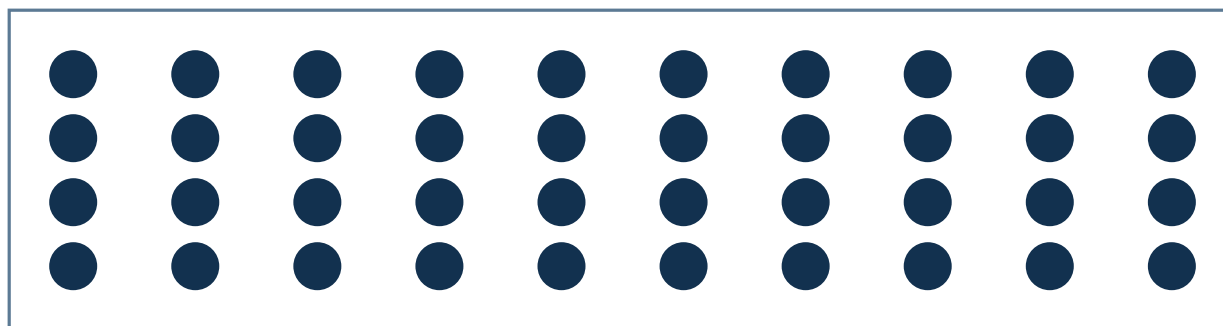


Diagram description: particles in a solid are drawn as evenly spaced dots arranged in tidy rows and columns, like a grid, showing a fixed, orderly pattern.

Key properties of solids

- Has a **fixed shape** -- it does not change shape unless a force is applied.
- Has a **fixed volume** -- it takes up the same amount of space no matter what container it is in.
- Particles are **tightly packed** and only vibrate in place.
- Generally **difficult to compress** (squeeze into a smaller space).

Everyday examples: ice cubes, a wooden chair, a rock, a coin, and a book.

3. Liquids

In a liquid, particles are still close together, but they have more energy than in a solid. They can slide past one another, which is why liquids can flow and change shape.

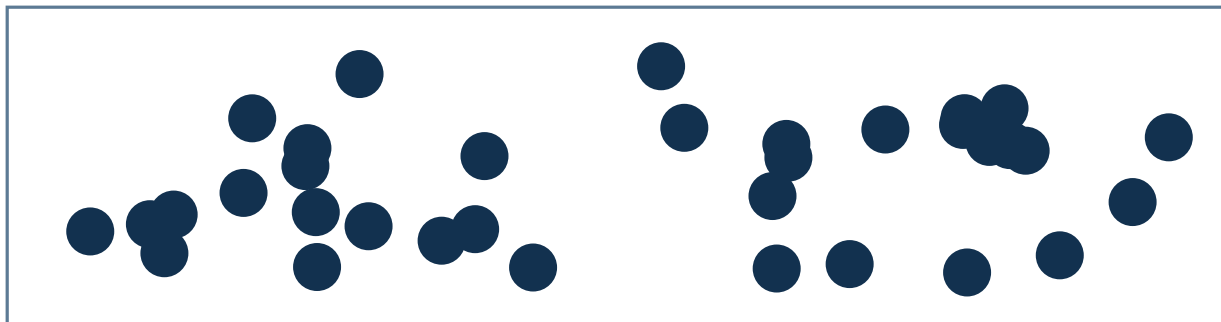


Diagram description: particles in a liquid are drawn as dots that are close together but scattered in an irregular, flowing arrangement rather than a neat grid, showing that they can move around each other.

Key properties of liquids

- Has **no fixed shape** -- it takes the shape of its container.
- Has a **fixed volume** -- the amount of liquid stays the same even if the container changes.
- Particles can **slide past each other**, allowing the liquid to flow.
- Generally **difficult to compress**, though slightly easier than a solid.

Everyday examples: water, milk, cooking oil, and juice.

4. Gases

In a gas, particles have the most energy of the three states. They move quickly in every direction and spread far apart, filling all the space available to them.

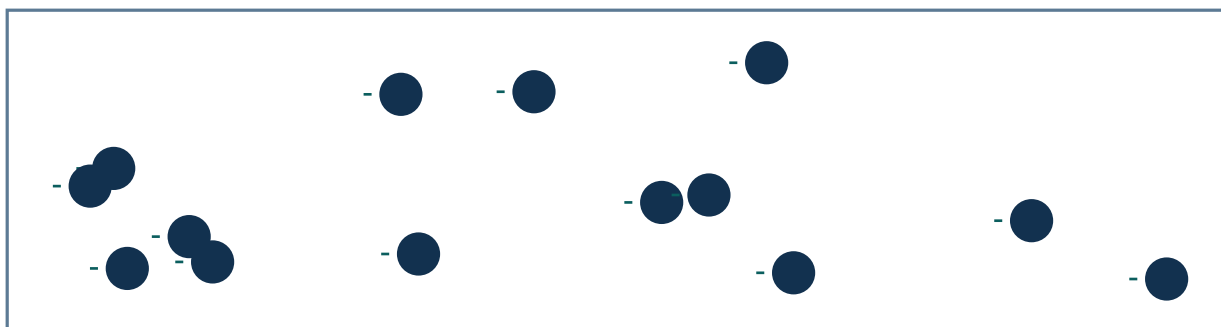


Diagram description: particles in a gas are drawn as widely spaced dots, each with a short line beside it representing fast, free movement in many directions.

Key properties of gases

- Has **no fixed shape** and **no fixed volume** -- it spreads out to fill its container completely.

- Particles move **quickly and freely**, with large spaces between them.
- **Easy to compress** because there is so much empty space between particles.

Everyday examples: the air around you, helium in a balloon, and steam rising from a hot drink.

5. Comparing the Three States

The table below summarizes the key differences. Row and column headers are labeled with text so the comparison does not depend on color alone.

Property	Solid	Liquid	Gas
Shape	Fixed	Takes shape of container	Takes shape of container
Volume	Fixed	Fixed	Fills available space
Particle spacing	Very close	Close	Far apart
Particle movement	Vibrate in place	Slide past each other	Move quickly and freely
Compressibility	Very low	Low	High

6. Changing From One State to Another

Adding or removing heat energy can change matter from one state to another. These changes are called **changes of state**. The names below describe the direction of each change.

Change	From	To	What happens to energy
Melting	Solid	Liquid	Energy is added; particles move faster and spread out.
Freezing	Liquid	Solid	Energy is removed; particles slow down and lock into place.
Evaporating / Boiling	Liquid	Gas	Energy is added; particles move fast enough to escape and spread far apart.
Condensing	Gas	Liquid	Energy is removed; particles slow down and move closer together.

Example: an ice cube (solid) left on a counter absorbs heat from the air. It melts into water (liquid). If that water is heated on a stove until it boils, some of it evaporates into water vapor (gas).

7. Everyday Examples

Water is one of the few substances people commonly see in all three states. The table below lists a few examples for each state.

Solid	Liquid	Gas
Ice	Water	Water vapor / steam
Wood	Cooking oil	Helium (balloon gas)
Rock	Milk	Air (oxygen, nitrogen)
Metal spoon	Juice	Carbon dioxide in soda

Try it yourself: look around the room you are in right now and name one solid, one liquid, and one gas you can find nearby.

8. Quick Check Questions

Answer each question in your own words or out loud. Check your answers in the next section.

Question 1

Which state of matter has a fixed shape and a fixed volume?

Question 2

Why can a liquid change shape but a gas can spread out to fill any container?

Question 3

What is it called when a solid changes into a liquid? What about liquid into gas?

Question 4

Name one example each of a solid, a liquid, and a gas that you might find in a kitchen.

Question 5

If you remove heat energy from a gas, what state might it eventually become?

9. Answer Key

Answer 1

A solid. It keeps its shape and volume because its particles are tightly packed and only vibrate in place.

Answer 2

A liquid's particles can slide past each other, so it flows and takes the shape of its container, but stays the same amount (volume). A gas's particles move quickly and have large gaps between them, so the gas spreads out to completely fill whatever space it is in.

Answer 3

Solid to liquid is called melting. Liquid to gas is called evaporating (or boiling, when it happens quickly throughout the liquid).

Answer 4

Example answer: ice (solid), milk (liquid), steam from a kettle (gas). Your own answer may differ and still be correct.

Answer 5

It would likely condense into a liquid, and could freeze into a solid if even more heat energy were removed.

Summary: what to remember

Matter is made of particles and commonly exists as a solid, liquid, or gas. Solids hold a fixed shape and volume because their particles are tightly packed. Liquids keep a fixed volume but flow to fit their container because their particles can slide past each other. Gases have no fixed shape or volume because their particles move quickly and spread far apart. Adding or removing heat energy can change matter from one state to another through melting, freezing, evaporating, and condensing.